

Chapter 6: Measurement of Physical Quantities

Textbook Exercise Solutions & Answers



1. Write answers to the following questions in your own words:

(a) Why is the weight of the same object different on different planets?

The weight of the same object is different on different planets because weight depends on the gravitational force of the planet and gravitational force on every planet is different so the weight also differs from planet to planet.

(b) What precautions will you take to make accurate measurements in day-to-day affairs?

Following precautions will be taken to make accurate measurements in day to day affairs:

1. The balance should carry the stamp of standardisation by the department of weights and measures.
2. Balance should be stable and the pointer of the balance should be upright.
3. The underside of the pan should not be tampered with any other weight or layers of other metal.
4. The weight should be made of metal.

(c) What is the difference between mass and weight?

Mass	Weight
1. The amount of matter present in a substance is called mass.	1. The gravitational force that acts on this mass is called its weight.
2. It is a scalar quantity.	2. It is a vector quantity.
3. Mass remains same everywhere on the earth.	3. Weight changes from place to place on the ' earth.

2. Who is my companion?

Column 'A' Column 'B' 1. Velocity a. Litre 2. Area b. Kilogram 3. Volume c. metre/second 4. Mass d. kilogram/cubic metre 5. Density e. square metre

Column 'A'	Column 'B'
1. Velocity	c. metre/second
2. Area	e. square metre
3. Volume	a. Litre
4. Mass	b. Kilogram
5. Density	d. kilogram/cubic metre

3. Explain giving examples.

(a) Scalar quantity

A quantity that can be completely expressed by its magnitude alone is called a scalar quantity, e.g. length, breadth, area, mass, temperature, density, time, work. In all these examples a value with a unit is used to express quantities. Thus, we say that the length of a tunnel is 2 km.

(b) Vector quantity

The quantity that is expressed completely only when magnitude and direction are both given is called a vector quantity. Displacement, velocity are vector quantities e.g. a displacement of 20 km towards north, the aeroplane flying at a velocity of 500 km/hr towards Mumbai.

4. Explain, giving examples, the errors that occur while making measurements.

Major causes of errors in measurement are:

1. Not using the appropriate device.
2. Not using the device properly.

Example: While buying things at grocery shops and the vegetable market remember to look out for the following:

- The balance should carry the stamp of standardisation by the department of weights and measures.
- Balance should be stable. The pointer of the balance should be upright.
- No changes should be done on the underside of the pan of the balance.
- Weight should be made up of metals and it has to be standardized.

5. Give reasons:

(a) It is not proper to measure quantities by using body parts as units.

It is not proper to measure quantities by using body parts as units because length of the body parts changes from person to person, so accurate measurement can not be found with the body parts as units.

(b) It is necessary to get the weights and measures standardized at regular intervals.

It is necessary to get the weights and measures standardized at regular intervals because of that accurate weight and measurement can be taken. Errors will be avoided.

6. Explain the need for accurate measurement and the devices to be used for that.

1. Measurement of substances that are precious or of great importance and used in very small quantities is done meticulously and accurately.
2. Due to advancements in technology, devices that measure very small magnitudes of quantities like distance, mass, time and temperature are available now.
3. e.g. distance and time in connection with very important sports competitions, mass of gold, body temperature etc.
4. Devices to be used for this are thermometer, digital balance (Analytical balance) etc.